for vani company website

Bhai, Vaani website ke experience ke according feedback **positive but genuine** rakhte hain. Aisa lagega ki tune website actually explore ki hai, sirf generic appreciation nahi likhi.

Ye 3 answers directly form mein paste kar sakta hai:

1. What stood out?

What stood out to me was the AI-powered agent-building experience. Instead of building an agent completely from scratch, I could describe the agent in natural language, including whom it should call, what it should say, and what outcome it should achieve. This makes agent creation more accessible and saves development time.

2. What confused you?

Initially, I wanted more clarity on how much control I would have over the agent after it was generated. It would be helpful to understand how to customize its behaviour, conversation flow, instructions, and responses, especially for more complex use cases.

3. What would you change?

I would add a more interactive step-by-step setup process with a live preview or testing environment. Users should be able to test conversations, modify the agent's instructions, and see how it responds before deploying it. This would make the overall experience more intuitive and give users greater confidence in the agent they create.

Meri suggestion: Ye feedback professional hai, constructive hai, aur Vaani ke AI agent-building experience se directly related hai.

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meri in 3 answer me se sabse best konsi h ?

meri Three important prompts konsi h ye chats me ? jo bhot imp tha for understanding and implementaition for this assignment

and 4. AI ki ek mistake aur ek overruled decision kya likhna hai? A. One AI mistake you caught
. One decision where you overruled AI

ye iske answer dena please acche se

3:11 PM

Tum reflection me yeh likh sakte ho. Yeh tumhari actual project journey se match karta hai:

Three important prompts

1. **“Pehle mera existing project inspect karo; code mat karo. Batao assignment me kya implemented hai aur kya missing hai.”**

Yeh important tha because isse project blindly rewrite nahi hua. Pehle folder structure, backend, frontend, PDFs, retrieval, scheduler aur existing code samjha gaya. Is prompt se minimum-change implementation plan bana.

2. **“PDF filename par trust mat karo. Har PDF ke andar actual arXiv ID aur version verify karo, aur SHA-256 manifest banao.”**

Yeh important tha because assignment me exact paper versions mandatory the. Isse Moshi v2 file identify hui aur v1 corpus ke liye use hui. Manifest ne paper version, page count aur checksum ka reproducible evidence diya.

3. **“Citation verification me citation ID hona enough nahi hai; factual claim evidence se actually supported bhi hona chahiye.”**

Yeh most important safety prompt tha. Isse system sirf fake/invalid citation reject nahi karta, balki unsupported claims, altered numbers, aur weakly supported answers ko safe refusal me convert karta hai.

A. One AI mistake I caught

AI mistake: Initial retrieval logic me generic word model ko paper-title matching ke liye use kiya gaya tha.

What happened: Question tha: “What is the base Transformer model dimension?” Generic word model ki wajah se system Moshi paper ko incorrectly prioritize kar raha tha aur 4096 answer de raha tha. Correct Transformer value d_model = 512 hai.

How I caught it: Maine output ka citation inspect kiya. Citation Attention Is All You Need paper ke instead Moshi paper ko point kar rahi thi. Maine question, retrieved chunks, paper metadata aur original paper evidence compare kiya.

Fix: Generic words (model, paper, time, etc.) ko title-anchor matching se remove kiya. Sirf distinctive title terms title expansion trigger karte hain. Is bug ke liye regression test bhi add hua.

B. One decision where I overruled AI

My decision: Maine AI ko real Groq rate limit force karke HTTP 429 generate karne nahi diya.

Reason: AI/provider testing ke liye multiple requests bhej kar real Groq quota exhaust kar sakta tha, but yeh unnecessary cost, account quota consumption aur provider limits ke against risky hota. Assignment ka purpose behavior demonstrate karna tha, provider ko stress karna nahi.

What I chose instead: Maine real Groq ke normal baseline measurements liye—latency, TTFT, token usage aur available response headers. Phir 429, Retry-After, failures, daily budget exhaustion aur heavy-load situations ko local configurable stand-in service par test kiya.

Why this was my contribution: Yeh engineering judgment tha: real evidence aur safe controlled simulation ko separate rakhna, instead of fabricated results ya unnecessary live-provider load create karna.

3:11 PM

meri Three important prompts konsi h ye chats me ? jo bhot imp tha for an implementaition like this **Handling multiple concurrent requests sytem architecure**

mera contribution kaise explain karna hai?

mera contribution kya tha is project me ? assigment complete krne me ?

3:17 PM

Haan — concurrency/system architecture wala prompt definitely include karo. Reflection me yeh likho:

Three important prompts

1. **“Design a simple single-process system to handle multiple concurrent requests using a bounded queue, worker pool, semaphore, deadlines, RPM/TPM limits and daily quota.”**

Yeh prompt important tha because Relay sirf normal chatbot nahi raha. Isse system architecture bana jahan multiple users ek saath request bhej sakte hain, lekin backend overload nahi hota. Queue pending requests ko control karti hai, workers requests process karte hain, semaphore simultaneous model calls limit karta hai, aur deadlines expired request par unnecessary work rok deti hain.

2. **“Verify exact arXiv PDF versions inside the PDFs and create a SHA-256 manifest before ingestion.”**

Isse correct research corpus ensure hua. Filename par trust nahi kiya gaya. Moshi v2 file detect hui aur assignment-required Moshi v1 hi index hui.

3. **“Citation integrity aur factual claim support ko alag verify karo; unsupported claim ko verified answer ke roop me return mat karna.”**

Isse RAG system safer bana. System check karta hai ki citation ID valid hai aur cited passage actual claim ko support bhi karta hai.

Mera contribution kya tha?

Tum apna contribution is tarah explain kar sakte ho:

Mera main contribution project ke engineering decisions aur validation me tha. Maine assignment brief ke according single-process architecture choose ki, jisme FastAPI backend, bounded queue, workers, semaphore, request deadlines, RPM/TPM quota limits aur daily budget fallback hai. Maine decide kiya ki project me unnecessary database, Redis, Kafka ya microservices add nahi karne hain.

Maine exact research PDFs/version verification par focus kiya, wrong Moshi version ko corpus se exclude karwaya, aur SHA-256 manifest use kiya. Maine RAG answer ke liye citations ko sirf display feature nahi rakha; citation integrity aur factual claim support dono verify karwaye.

Testing ke time maine retrieval results inspect kiye aur Transformer dimension question ka incorrect Moshi-based answer catch kiya. Iske baad generic model title-expansion bug fix hua aur regression test add hua.

Maine real Groq service ko unnecessarily rate-limit hit karne ke liye stress test karne ke decision ko reject kiya. Instead, normal Groq measurements liye aur high-load, 429, Retry-After, failures aur daily-budget exhaustion ko controllable local stand-in model se test kiya.

Finally, maine raw experiment data, CSV schema, load-test results, A/B/C retrieval evaluation aur submission evidence ko review karke ensure kiya ki fabricated measurements na hon.

Short version:

My contribution was defining the architecture, validating the correct research corpus, reviewing retrieval errors, deciding the safety policy, selecting controlled stand-in testing instead of forcing Groq limits, and ensuring that final results were evidence-based rather than fabricated.

3:17 PM

Work locallyLocal

Approve for me

Select model5.6 TerraMediumNoneMinimalLightMediumHighExtra HighMaxUltraPersistent

IDE context

Bhai, tumhari The Relay project journey mein mere according ye 3 sabse important prompts the. Inhone directly tumhari architecture, implementation aur testing decisions shape kiye.

Main inhe importance ke order mein rakh raha hoon. Wording ko report-ready banaya hai; ye original chats ke verbatim quotes nahi, balki unka reconstructed version hai.

1. Handling Multiple Concurrent Requests – System Architecture

Most important

Prompt:

Design a production-ready system architecture for my AI research assistant that can handle multiple concurrent user requests. Explain how to manage request queues, worker pools, concurrency limits, semaphores, RPM/TPM rate limits, token budgets, retries, timeouts, backpressure and API failures. Explain the complete request lifecycle, bottlenecks, trade-offs and how to handle traffic spikes without overwhelming the LLM provider.

Is prompt se kya implement hua?

- FastAPI admission control.
- Bounded request queue.
- Worker pool and semaphore.
- RPM/TPM and daily quota scheduler.
- Retry, exponential backoff and timeout.
- Groq API request management.

Why this prompt mattered: Isne simple chatbot ko production-oriented concurrent AI system mein convert karne ki direction di.

2. RAG Architecture – Three Research Papers with Grounded Answers

Prompt:

Design and implement a RAG-based AI research assistant that answers questions exclusively from three provided research papers. Explain PDF ingestion, text extraction, chunking, embeddings, BM25 retrieval, vector search, evidence checking, LLM generation and citation verification. The chatbot must provide accurate paper references and page numbers, and refuse to answer when sufficient supporting evidence is unavailable. Explain the complete pipeline and implementation using Python.

Is prompt se kya implement hua?

- PDF ingestion and chunking.

- BM25 retrieval.
- A/B/C retrieval configurations.
- Evidence sufficiency checking.
- Citation integrity verification.
- Claim-support verification.
- Safe refusal mechanism.

Why this prompt mattered: Isne chatbot ka core intelligence aur hallucination-prevention mechanism define kiya.

3. Load Testing, Stand-in Model and Performance Evaluation

Prompt:

Design a comprehensive testing and evaluation strategy for my AI research assistant. Create a stand-in LLM service that simulates realistic latency, streaming, HTTP 429, transient failures, retries and daily quota exhaustion. Explain how to test steady traffic, burst traffic and sustained overload. Measure TTFT, total latency, throughput, queue waiting time, token consumption, retries and error rates. Compare different retrieval configurations using manually graded questions and generate reproducible results.csv and performance reports.

Is prompt se kya implement hua?

- Local stand-in model.
- Five load-testing scenarios.
- Groq baseline measurements.
- A/B/C retrieval comparison.
- 48 reviewed evaluation records.
- Latency and token metrics.
- results.csv and reporting scripts.

Why this prompt mattered: Isne ensure kiya ki system sirf work karne ka claim na kare, balki measurable experiments se apni reliability prove kare.

Final selection for your assignment reflection

My three most important AI prompts

1. Production-ready system architecture for handling multiple concurrent requests.

Designed the queue, workers, concurrency controls and rate-limit management.

2. Grounded RAG implementation using three research papers with ci

…

Bhai, tumhari complete project journey dekhkar main ye final version recommend karunga. Ismein **Vaani website feedback**, tumhare **3 most important prompts**, **AI mistake**, **AI ko overrule karne ka decision**, aur **tumhara actual contribution** sab cover hain.

Ek important distinction: prompts ki wording reconstructed hai, original chats ke exact verbatim quotes nahi. Reflection mein is baat ko honestly mention karna.

1. Vaani website feedback — best answers

Mere according teeno answers ka combination best hai: first mein product ki strength, second mein genuine confusion aur third mein practical improvement.

1. What stood out?

What stood out to me was the AI-powered agent-building experience. Instead of building an agent completely from scratch, I could describe the agent in natural language, including whom it should call, what it should say, and what outcome it should achieve. This makes agent creation more accessible and reduces the effort required to get started.

2. What confused you?

Initially, I wanted more clarity on how much control I would have over the agent after it was generated. It would be helpful to understand how to customize its behaviour, conversation flow, instructions, and responses, especially for more complex use cases.

3. What would you change?

I would add an interactive testing environment with a live preview. Users should be able to test conversations, modify the agent's instructions, and see how the agent responds before deploying it. This would make the overall experience more intuitive and give users greater confidence in the agent they create.

2. Three most important prompts from The Relay

Ye teen prompts select karo. Ye project ke three major engineering dimensions represent karte hain.

Prompt 1: Handling Multiple Concurrent Requests — System Architecture

"Design a system architecture for handling multiple concurrent users and requests in The Relay. The system should handle traffic bursts, multiple requests executing simultaneously, Groq RPM/TPM/daily limits, failures, timeouts and user cancellations. Explain the queue, workers, concurrency control, rate limiting, retries, deadlines and trade-offs."

Why it was important:

This helped shape the request-processing architecture, including the FastAPI admission controller, bounded request queue, worker pool, semaphore-based concurrency control, RPM/TPM limits, daily quota management, and retry and timeout handling.

Prompt 2: RAG Architecture — Evidence-Grounded Answers

"Design a RAG-based research assistant that answers questions exclusively using the three approved research papers. Explain PDF ingestion, chunking, BM25 retrieval, evidence sufficiency checking, citation generation, citation integrity and claim-support verification. The assistant must refuse to answer when sufficient supporting evidence is unavailable."

Why it was important:

This defined the core AI pipeline: processing the approved papers, retrieving relevant passages, generating page-level citations, checking factual support, and refusing unsupported questions.

Prompt 3: Load Testing and Failure Simulation

"Design a testing strategy for The Relay that evaluates steady traffic, burst traffic, sustained overload, reduced RPM limits and daily budget exhaustion. Create a local stand-in model to simulate latency, streaming, HTTP 429/503 failures and retries. Measure TTFT, total latency, token usage, retries and answer correctness."

Why it was important:

This guided the stand-in service, five load-testing scenarios, retrieval configuration comparisons, performance measurements and reproducible evaluation reports.

Overall: These three prompts helped me understand and guide System Design, AI Engineering and Reliability Engineering in the project.

3. One AI mistake I caught

One AI mistake I caught

During retrieval testing, I noticed that a question about the base Transformer model dimension was incorrectly retrieving the Moshi paper and returning an unsupported answer of 4096 instead of the correct value, d_model = 512, from the *Attention Is All You Need* paper.

I investigated the retrieved results and checked the citation against the original research paper. I identified that the generic word "model" was incorrectly influencing title-based matching.

I asked for the retrieval logic to be corrected so that only distinctive paper-title terms would trigger title matching, and a regression test was added to prevent the issue from recurring.

This taught me not to trust AI-generated answers just because they contain a citation. The supporting evidence must actually match the question and support the answer.

4. One decision where I overruled AI

One decision where I overruled AI

I decided not to deliberately exhaust or force rate limits on the real Groq API just to demonstrate HTTP 429 behaviour.

Doing so could consume unnecessary API quota and create avoidable provider-side rate-limit issues. Instead, I used the real Groq service for normal baseline measurements and used a configurable local stand-in service to simulate HTTP 429 responses, Retry-After behaviour, transient failures, latency and daily budget exhaustion.

This allowed me to test failure-handling and retry mechanisms in a controlled and reproducible environment without unnecessarily stressing the real provider.

I considered this an important engineering decision because it separated real provider measurements from simulated stress-test results and avoided presenting simulated results as actual Groq performance.

5. My contribution to The Relay

Ye section sabse important hai. Interviewer ko samajhna chahiye ki AI coding assistance use karne ke bawajood tumne engineering direction, decisions aur validation mein kya role play kiya.

What was distinctly my contribution?

My primary contribution to The Relay was understanding the assignment requirements, defining the engineering direction, guiding implementation, and critically validating the system.

I focused on designing a reliable RAG-based research assistant that answers questions exclusively using three approved research papers, provides supporting citations, and refuses to answer when sufficient evidence is unavailable.

One of my major contributions was exploring and guiding the architecture for handling multiple concurrent requests. I focused on the request queue, worker pool, semaphore-based concurrency control, RPM/TPM limits, daily quotas, retries, deadlines and failure handling.

I also worked on the RAG design, including PDF processing, retrieval strategies, evidence verification, citation integrity and claim-support validation.

During testing, I reviewed retrieval outputs, investigated incorrect answers, and identified a retrieval issue involving generic title matching. I also focused on evaluating the system under five load scenarios and comparing retrieval configurations using latency, token usage and answer-quality metrics.

Another important contribution was making practical engineering decisions, such as using a local stand-in service for controlled failure and load testing instead of unnecessarily exhausting the real Groq API quota.

I used AI coding assistance to accelerate development, but I guided the technical approach, reviewed generated changes, investigated issues, requested corrections and checked the implementation against the assignment requirements.

Overall, my contribution was in problem understanding, system architecture, RAG pipeline design, engineering decisions, testing strategy, debugging and technical validation.

Meri final recommendation

Assignment reflection mein in points ko priority dena:

Reflection question	**Your strongest answer**
Three important prompts	Concurrent architecture, grounded RAG, load testing
AI mistake caught	Incorrect retrieval due to generic title matching
Decision overruled	Not forcing real Groq rate limits; using stand-in

Your contribution	Architecture decisions, implementation guidance, debugging and validation

Ek chhoti si caution: AI mistake wale example mein 4096 versus 512 aur regression test ka detail tabhi submit karna jab tum apne actual project logs/test evidence mein ise confirm kar sako. Isi tarah contribution mein wahi implementation claim karna jise tum explain aur defend kar sakte ho.

Mere hisaab se tumhari strongest story ye hai: **You didn't just build a chatbot; you helped shape, challenge, test and validate a reliable AI research system.**

ye kya mila aur kya bacha h ?

Aapke diye genuine notes se submission reflection complete kar diya.

Completed:

- [Vaani feedback](E:\vani ai assignment\Backend\submission\VAANI_FEEDBACK_TEMPLATE.md)
    - Teacher AI, Memories, Knowledge vs Memories confusion, onboarding suggestion, freshness-monitoring idea.
- [Assistant conversation reflection](E:\vani ai assignment\Backend\submission\CONVERSATION_HISTORY_CHECKLIST.md)
    - 3 important prompts:
        1. Concurrent request architecture
        2. Evidence-grounded RAG
        3. Load testing/failure simulation
    - AI mistake caught
    - Overruled Groq-limit decision
    - Your distinct contribution
- [Final report](E:\vani ai assignment\Backend\submission\REPORT_FINAL.md)
    - Vaani feedback and your assistant-working section added.
- [Final eight-task status](E:\vani ai assignment\Backend\submission\FINAL_EIGHT_TASK_REPORT.md)
    - Task 6 is now marked complete.

Only final PDF remains blocked because the required PDF artifact-marker utility is unavailable in this environment. I have not falsely claimed that a two-page PDF was created or visually checked. If your college portal only needs the report file, the Markdown report content is ready; if it strictly requires PDF, export REPORT_FINAL.md locally through Word/Google Docs and check it is exactly two pages

…

Bhai, ab project submission ka kaafi important kaam complete ho gaya hai. ❤️ Main simple breakdown deta hoon ki **kya complete mila hai aur abhi kya pending hai.**

1. Kya complete ho gaya hai?

4 items updated / marked complete

File	Kya mila hai?
VAANI_FEEDBACK_TEMPLATE.md	Vaani website ke 3 feedback answers
CONVERSATION_HISTORY_CHECKLIST.md	3 important prompts, AI ki mistake, AI ko overrule karne ka decision aur tumhara contribution
REPORT_FINAL.md	Final report mein Vaani feedback aur assistant reflection add kar diye gaye
FINAL_EIGHT_TASK_REPORT.md	Task 6 ko complete mark kar diya gaya

Reflection mein ye content cover hai:

- Three important prompts: Concurrent request architecture, grounded RAG, load testing.
- AI mistake: Generic title matching ki wajah se incorrect retrieval.
- Overruled decision: Real Groq rate limit force nahi ki; stand-in service use ki.
- Your contribution: Architecture, implementation guidance, debugging, testing and validation.

2. Abhi kya bacha hai?

PDF pending

1. Final report ko PDF banana

REPORT_FINAL.md ready hai, lekin actual PDF create nahi hua.

Tumhe:

- Markdown report ko Word ya Google Docs mein export karna hai.
- PDF format mein save karna hai.
- Check karna hai ki report exactly 2 pages ki hai.
- PDF open karke formatting, tables aur text readability verify karni hai.

2. Actual conversation evidence verify karna

Written reflection complete hai. Lekin assignment mein agar actual ChatGPT conversation history, screenshots ya export maanga hai, toh woh evidence bhi attach karna hoga. Sirf reconstructed prompts ko actual chat transcript mat samajhna.

3. Final submission check

PDF aur supporting evidence ko assignment requirements ke against ek baar verify karna hai.

3. Final status

Reflection and written content

Complete

Vaani feedback

Complete

Three prompts and personal contribution

Complete

Final Markdown report

Ready

Two-page PDF

Pending

Actual conversation evidence

Verify

Main remaining task: PDF export, visual check and final evidence verification.

Meri advice: Ab Codex se poora project dobara modify mat karwana. Implementation audit aur written reflection ho chuke hain. Ab final PDF aur evidence par focus karo.

Agar tum REPORT_FINAL.md aur assignment brief upload kar do, toh main report ko assignment requirements ke against review karke bata sakta hoon ki submission mein kuch missing toh nahi hai.

1 selection

matlb apne chats ke ss bheju use ?

Haan bhai! ❤️‍🩹 **Apni actual ChatGPT chats ke screenshots dene hain**, taaki evaluator dekh sake ki tumne AI assistant ko kaise use kiya, guide kiya aur uske answers ko verify kiya.

Lekin saari chats ke screenshots bhejne ki zaroorat nahi hai. Important conversations select karo.

Kaunse screenshots lene hain?

Screenshot	**Kya capture karna hai?**
1. Concurrent request architecture	Tumhara prompt aur ChatGPT ka architecture explanation
2. RAG implementation	Three research papers, retrieval, citations aur refusal wala discussion
3. Load testing	Stand-in model, rate limits aur five testing scenarios wala discussion

4. AI mistake	Incorrect retrieval identify karne aur usko correct karwane wali chat
5. Overruled decision	Groq API ko forcefully rate-limit test na karne aur stand-in use karne wali discussion
6. Your contribution	Tumhare decisions, follow-up questions aur implementation review ke examples

Screenshots kaise lene hain?

1. Actual ChatGPT conversation open karo.
2. Relevant prompt aur uska response ek screenshot mein capture karo.
3. Jahan possible ho, tumhara follow-up question aur correction bhi include karo.
4. Screenshots ko edit karke conversation ka meaning change mat karna.
5. API keys, passwords ya personal information visible ho toh hide kar dena.

Meri recommendation: 6–10 relevant screenshots sufficient starting evidence ho sakte hain. Complete chat history export available hai toh woh bhi useful hai.

Tum screenshots yahan upload kar dena. Main check karke bata dunga:

- Kaunse screenshots assignment ke liye strongest hain.
- Kaunse duplicate ya unnecessary hain.
- Unhe kis order mein arrange karna hai.
- Final submission mein kis reflection point ke saath attach karna hai.

The Relay - final report

1. What I built and why

Relay is a single-process Python RAG assistant for exactly three verified arXiv papers. It extracts page-aware chunks, retrieves evidence, asks Groq for structured claims, then checks both citation identity and claim-to-evidence support before returning an answer. The design decision that shaped the system was ****verification before exposure****: an answer can be slower, but it must not be presented as grounded unless the cited passages support it.

2. The real service and my stand-in

I recorded eight modest real Groq calibration calls using `openai/gpt-oss-20b`: model-TTFT mean 612.2 ms, p50 678.1 ms, p95 732.1 ms; total-latency mean 625.9 ms, p50 689.1 ms, p95 743.0 ms. Each calibration response used 91 input and 77 output tokens. Returned headers reported an RPD value of 1000 and TPM value of 8000 for this active model. RPM, TPD, concurrent limit, and real-429 recovery were not measured, so I do not claim them.

The stand-in loads a calibration derived from those measurements: 639.1 ms base delay, 1.0 ms estimated token delay, 0.25 load-slowdown multiplier, configurable 503 failure, daily quota, and 429

plus Retry-After. I validated SSE pieces, 429/Retry-After, and 503 locally. It differs most from Groq because its generation quality and failure distribution are synthetic.

3. Results

Five stand-in scenarios were retained as raw rows. Repeated local attempts remain in the raw data, so duration/offered rate/peak queue cannot be reconstructed and are marked not measured.

Scenario	Requests	Total p50/p95 ms	Success %	Rate-limited %
Steady	17	825.8 / 861.1	23.5	76.5
Burst	25	1433.1 / 1549.1	40.0	60.0
Sustained overload	127	963.2 / 1204.0	10.2	89.8
RPM halved	26	808.0 / 831.9	38.5	46.2
Daily budget exhausted	34	38.6 / 1773.3	58.8	20.6

All figures and missing-data notes are reproducible from `results.csv`, `results_submission_clean.csv`, and `REQUIRED_TABLES.md`.

4. Where the time goes

The slowest trace was a C-setting stand-in request: 78,539.0 ms total while provider model work was 828.0 ms and retry count was zero. A slow Groq C run took 37,520.6 ms total while model work was 1,318.7 ms and provider TTFT was 1,029.9 ms. Therefore most observed delay occurred before verified response delivery, not after the provider's first token. Existing historical traces do not split queue, retrieval/model-load, and verification timestamps separately; that is a measurement limitation, not an invented attribution.

5. When a limit is hit

The scheduler admits work only within RPM, TPM, daily budget, queue capacity, concurrency, and deadline limits. Queue-full and expired requests are rejected. Per-minute 429/5xx responses use bounded exponential backoff with jitter and honor Retry-After. Under pressure it uses smaller retrieval while keeping safety verification. At daily exhaustion it makes ****no provider call**** and sends cited retrieved passages with an unavailability explanation.

Retrieval: quality versus cost

Setting	Correct cited %	Wrong %	Correctly refused %	Tokens/answer	TTFT p50/p95 ms
A: 225 words, top 3	62.5	12.5	25.0	1556.2	1062.1 / 1739.9
B: 500 words, top 5	68.8	6.2	25.0	2033.6	941.8 / 1233.8
C: 500 words, BM25 8 then MiniLM 4	62.5	12.5	25.0	1658.0	1029.9 / 1372.6

I select ****B-medium****. It was best on the same 16-question review: 11 correct cited, one wrong, and four correct refusals. C was a real bi-encoder experiment, not a claimed win. Its original score included a generic-title expansion bug and a structured-output refusal; the retrieval bug was fixed and regression-tested, but I did not rewrite its original result.

How this library will mislead a naive retriever

1. Shared generic words cause paper confusion: "base Transformer model dimension" originally pulled Moshi through `model`.
2. Ambiguous latency questions have several valid numbers: "What latency does Moshi report?" can

retrieve 80, 160, 240, or 640 ms in different contexts.
3. Short names are weak queries: "What is H-Net?" has only `h`/`net`, making formatting and ranking fragile.

6. Retries

Retries are limited to two transient 429/5xx failures, use Retry-After when supplied, then exponential backoff plus jitter. Retrying a daily-quota 429 would only delay the user (the stand-in returns Retry-After 86400), so it is classified as daily exhaustion and switched to retrieval-only output.

7. What the person sees

At 3pm daily budget exhaustion, the user receives an explicit message that the daily model budget is exhausted plus up to three cited retrieved passages. They do not receive an unverified synthesis.

8. What I am unsure about, left out, and learned

My least certain assumption is that eight short real calibration calls represent provider behavior under launch load; they do not. I deliberately did not add Redis/microservices or unverified frontend token streaming. I learned that citation existence and citation support are different checks, and that semantic rescoring can still worsen a small-paper corpus.

Your feedback on Vaani

The student explored Vaani and found Teacher AI and Memories especially useful: natural-language agent requirements, reviewable prompt changes, and retained call context reduce manual configuration and repeated customer information. Knowledge versus Memories and the Teacher AI readiness score were less clear. The suggested improvement is guided onboarding plus knowledge freshness/accuracy monitoring with version history and outdated-answer alerts.

How I worked with the assistant

The student used the assistant to explore concurrent-request architecture, evidence-grounded RAG, and failure/load testing. The student caught a retrieval bug in which a generic `model` title match pulled Moshi's 4096 dimension for a Transformer question; the bug was fixed and regression-tested. The student also chose not to force real Groq rate limits, using modest real measurements and controlled stand-in failure tests instead. The complete student reflection is retained in `CONVERSATION_HISTORY_CHECKLIST.md`.

…

Bhai, ab tumhara **updated final report mil gaya**. Is version mein pehle se zyada detailed information hai, aur Vaani feedback bhi tumhare actual exploration ke according update hua hai.

Maine content review kiya. Ye raha current status:

1. Report mein kya-kya cover ho gaya hai?

Content covered

Section	**Status**
What I built and why	Covered

Real Groq calibration and stand-in	Covered
Five load-testing scenarios	Covered
Latency analysis	Covered
Rate limits, retries and quota handling	Covered
Retrieval A/B/C comparison	Covered
AI mistake and correction	Covered
Overruled decision	Covered
Vaani website feedback	Covered
Your contribution and AI collaboration	Covered

2. Abhi kya bacha hai?

Mere hisaab se ye final four checks important hain:

1. **Actual ChatGPT conversation evidence:** Apni original chats ke screenshots ya export attach karo. Especially concurrent architecture, RAG, load testing, AI mistake aur Groq decision wali conversations.
2. **Exactly two-page PDF:** Report ko PDF mein export karke page count aur formatting check karo. Content detailed hai, isliye two-page requirement ko specially verify karna.
3. **Required supporting files:** results.csv, results_submission_clean.csv, REQUIRED_TABLES.md aur baaki required submission artifacts folder mein present aur readable hone chahiye.
4. **Final evidence verification:** Jo report mein claims hain, jaise 8 Groq calibration calls, 48 reviewed records aur regression test, unke supporting logs/results available hone chahiye.

3. Ek important observation

Report mein ek achhi engineering decision clearly explain hui hai:

B-medium retrieval configuration select kiya, kyunki 16-question evaluation mein 11 correct cited answers aur 4 correctly refused answers mile. C configuration ko artificially winner nahi dikhaya. Ye honest evaluation hai.

Ek aur strong point hai: tumne missing measurements ko *not measured* mark kiya hai. Ye fabricated performance numbers dene se kaafi better hai.

Final status

Submission nearly ready

Written content complete; final evidence and packaging checks remain.

Report content

Ready

Reflection

Ready

Actual ChatGPT screenshots

Pending verification

Two-page PDF

Pending

Final submission folder

Final check

Meri advice: Ab implementation ko unnecessarily change mat karna. Pehle REPORT_FINAL.md, original assignment PDF aur submission folder mujhe de do. Main final rubric-wise audit karke bataunga ki submit karne se pehle exactly kya missing hai.

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isme jo bola tha sab achieve hua na ?

Bhai, maine tumhara **final report assignment brief ke saath point-by-point compare kiya hai**. Honest answer: **project ke major engineering goals achieve hue hain, lekin assignment ki har requirement abhi fully satisfied ya documented nahi hai.**

Main tumhe clearly bata raha hoon ki kya complete hai aur kya fix karna zaroori hai.

1. Jo requirements achieve hui hain

Covered in the report

Assignment requirement	**Status**
Python-based RAG assistant	Covered
Exactly 3 specified research papers	Covered
Evidence-based answers and citations	Covered
Unsupported questions par refusal	Covered
Real Groq API calibration	8 calls recorded

Local stand-in model	Covered
Slow responses, failures, 429 and 503 simulation	Covered
Three or more load scenarios	5 scenarios
Three retrieval configurations	A, B, C
At least 15 evaluation questions	16 per setting
Vaani feedback	Covered
AI mistake and overruled decision	Covered

Ye requirements assignment ke pipeline, model, load testing, retrieval aur reflection sections se match karti hain.

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2. Jo requirements abhi incomplete hain

Needs attention

1. Required Tables 1, 2a, 2b and 3

Assignment mein specific columns ke saath complete tables mandatory hain.

Tumhare pasted report mein:

- Table 1 ke saare provider limits, discovery method aur confidence columns nahi dikh rahe.
- Table 2a mein load scenarios ke TTFT p50/p95/p99, duration, offered rate aur total latency percentiles missing hain.
- Table 2b mein timed-out/given-up percentage, retries per request, peak queue length, tokens per answer aur cost per 1,000 answers missing hain.
- Table 3 mein total-time p50/p95 missing hai.

Tumhare REQUIRED_TABLES.md mein ye values hon, tab bhi ensure karo ki required tables final two-page report mein bhi copy ki gayi hon. Assignment explicitly ye maangta hai.

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2. Where the time goes

Assignment specifically kehta hai ki ek slow request ko teen parts mein split karo:

- Waiting time
- Working time

- Retrying time

Aur batao delay first token se pehle hua ya baad mein.

Tumhare report mein khud likha hai ki historical traces mein queue, retrieval/model-load aur verification ka separate breakdown available nahi hai.

Isliye ye requirement **fully achieved nahi hai**. Provider model work aur total latency ka comparison useful hai, lekin required three-part breakdown nahi hai.

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3. Real Groq limits

Tumne eight real calls se latency aur kuch response headers measure kiye. Lekin:

- RPM: Not measured
- TPD: Not measured
- Concurrent request limit: Not measured
- Real HTTP 429 response and recovery: Not measured

Inko honestly not measured likhna sahi hai. Lekin assignment documentation aur provider response se limits investigate karne ko bhi kehta hai. Isliye Table 1 ko complete karna abhi important hai.

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4. Retry made things worse

Tumne daily-quota 429 ko retry nahi karne ka sensible decision explain kiya hai. Lekin assignment ek specific case bhi maangta hai jahan retry karne se situation worse hui.

Report mein ise example ke roop mein clearly explain karo aur batao ki delay ya unnecessary waiting kaise badhti.

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5. Actual ChatGPT conversation history

Tumhare three prompts aur reflection ka written summary hai. Lekin actual conversation export/screenshots bhi submission mein attach karne hain, ya tool export nahi kar sakta toh actual prompts.

Reconstructed prompts ko verbatim original conversation mat present karna.

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6. Exactly two-page PDF

Final report ka PDF abhi export aur verify karna baaki hai. Assignment maximum two pages of writing kehta hai, code, raw results aur conversation history uske additional attachments hain.

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3. Final achievement status

Major project work complete

Based on the report you shared, not a fresh source-code inspection.

Core RAG and architecture

Covered

Groq and stand-in experiments

Covered

Five load scenarios

Covered

Complete required result tables

Verify / complete

Slow request breakdown

Incomplete

Retry-worsened case

Clarify

Conversation evidence

Pending

Two-page PDF

Pending

4. Meri recommendation

Bhai, project ko dobara build karne ki zaroorat nahi lag rahi. Ab mainly report aur evidence ki completeness fix karni hai.

Sabse important: mujhe REQUIRED_TABLES.md, results.csv aur REPORT_FINAL.md bhi upload kar do. Main inhe assignment brief ke saath verify karke exact bataunga ki kaunse numbers aur sections missing hain, aur kya final submission mein confidently submit kiya ja sakta hai.

Main abhi tumhe 100% complete bolkar mislead nahi karunga, lekin tumhara major engineering work kaafi achhe stage par hai. ❤️